ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 84 cm²	12×7
2	**(d)** 30 cm²	$\frac{1}{2} \times 10 \times 6$
3	**(d)** 36 cm²	base × height $= 9 \times 4$
4	**(b)** 35 cm²	$\frac{1}{2}(8+6) \times 5$
5	**(c)** 40 cm²	$\frac{1}{2} \times 10 \times 8$
6	**(c)** 25 cm²	Side $= 5$ cm; area $= 5^2$
7	**(b)** 120 cm²	15×8
8	**(b)** 42 cm²	$\frac{1}{2} \times 12 \times 7$
9	**(c)** 66 cm²	base × height $= 11 \times 6$
10	**(c)** 42 cm²	$\frac{1}{2}(10+4) \times 6$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $= l \times b = 25 \times 12$ (1) $= 300$ m² (1).	1+1
12	Area $= \frac{1}{2}bh = \frac{1}{2} \times 18 \times 10$ (1) $= 90$ cm² (1).	1+1
13	Area $= \frac{1}{2}d_1d_2 = \frac{1}{2} \times 22 \times 10$ (1) $= 110$ cm² (1).	1+1
14	Area $= l \times b = 30 \times 8$ (1) $= 240$ m² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= bh = 126$ cm² (2); triangle $= \frac{1}{2} \times 126 = 63$ cm² (1).	2+1
16	Area $= \frac{1}{2}(a+b)h = \frac{1}{2}(13+7) \times 9$ (2) $= 90$ cm² (1).	2+1
17	Area $= 8 \times 5 = 40$ m² (2); cost $= 40 \times 40 =$ ₹ 1600 (1).	2+1
	OR: Area $= bh = 112$ cm² (2); triangle $= \frac{1}{2} \times 112 = 56$ cm² (1).	2+1
18	Area $= \frac{1}{2}(a+b)h = \frac{1}{2}(16+10) \times 6$ (2) $= 78$ cm² (1).	2+1
	OR: Area $= 10 \times 6 = 60$ m² (2); cost $= 60 \times 30 =$ ₹ 1800 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Side $= 24 \div 4 = 6$ m (1); area $= 6^2 = 36$ m² (2); tiles $= 36$ (2).	1+2+2
	OR: Area $= \frac{1}{2} \times 16 \times 12 = 96$ cm² (3); each triangle $= \frac{1}{4}$ of it $= 24$ cm² (2).	3+2
20	Side $= 32 \div 4 = 8$ m (1); area $= 8^2 = 64$ m² (2); tiles $= 64$ (2).	1+2+2
	OR: Area $= \frac{1}{2} \times 20 \times 14 = 140$ cm² (3); each triangle $= \frac{1}{4}$ of it $= 35$ cm² (2).	3+2

Mistakes to avoid:

- Using the slant side instead of the perpendicular height in the area of a parallelogram or triangle.
- Forgetting the factor $\frac{1}{2}$ in the area of a triangle, trapezium or rhombus.
- Mixing units: multiplying a length in metres by one in centimetres; convert first.
- Writing area in cm instead of cm^2, or converting m^2 to cm^2 by multiplying by 100 (it is 10 000).
- Using the sum of the non-parallel sides for a trapezium; only the **parallel** sides enter the formula.
- Adding instead of subtracting when an L-shaped region is obtained by removing a corner.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 54 cm²	$\frac{1}{2} \times 12 \times 9$
2	**(a)** 49 cm²	Side $= 7$ cm; area $= 7^2$
3	**(d)** 54 cm²	9×6
4	**(d)** 36 cm²	$\frac{1}{2} \times 8 \times 9$
5	**(c)** 65 cm²	base × height $= 13 \times 5$
6	**(a)** 40 cm²	$\frac{1}{2}(12 + 8) \times 4$
7	**(d)** 42 cm²	$\frac{1}{2} \times 14 \times 6$
8	**(a)** 81 cm²	Side $= 9$ cm; area $= 9^2$
9	**(a)** 90 cm²	18×5
10	**(c)** 35 cm²	$\frac{1}{2} \times 14 \times 5$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $= \frac{1}{2}bh = \frac{1}{2} \times 24 \times 7$ (1) $= 84$ cm² (1).	1+1
12	Area $= \frac{1}{2}d_1d_2 = \frac{1}{2} \times 26 \times 8$ (1) $= 104$ cm² (1).	1+1
13	Area $= l \times b = 16 \times 14$ (1) $= 224$ m² (1).	1+1
14	Area $= \frac{1}{2}bh = \frac{1}{2} \times 15 \times 12$ (1) $= 90$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= bh = 132$ cm² (2); triangle $= \frac{1}{2} \times 132 = 66$ cm² (1).	2+1
16	Area $= \frac{1}{2}(a + b)h = \frac{1}{2}(21 + 9) \times 8$ (2) $= 120$ cm² (1).	2+1
17	Area $= 12 \times 7 = 84$ m² (2); cost $= 84 \times 25 =$ ₹ 2100 (1).	2+1
	OR: Area $= bh = 108$ cm² (2); triangle $= \frac{1}{2} \times 108 = 54$ cm² (1).	2+1
18	Area $= \frac{1}{2}(a + b)h = \frac{1}{2}(18 + 12) \times 5$ (2) $= 75$ cm² (1).	2+1
	OR: Area $= 9 \times 8 = 72$ m² (2); cost $= 72 \times 50 =$ ₹ 3600 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Side $= 40 \div 4 = 10$ m (1); area $= 10^2 = 100$ m² (2); tiles $= 100$ (2).	1+2+2
	OR: Area $= \frac{1}{2} \times 24 \times 10 = 120$ cm² (3); each triangle $= \frac{1}{4}$ of it $= 30$ cm² (2).	3+2
20	Side $= 52 \div 4 = 13$ m (1); area $= 13^2 = 169$ m² (2); tiles $= 169$ (2).	1+2+2
	OR: Area $= \frac{1}{2} \times 18 \times 16 = 144$ cm² (3); each triangle $= \frac{1}{4}$ of it $= 36$ cm² (2).	3+2

Mistakes to avoid:

- Using the slant side instead of the perpendicular height in the area of a parallelogram or triangle.
- Forgetting the factor $\frac{1}{2}$ in the area of a triangle, trapezium or rhombus.
- Mixing units: multiplying a length in metres by one in centimetres; convert first.
- Writing area in cm instead of cm^2, or converting m^2 to cm^2 by multiplying by 100 (it is 10 000).
- Using the sum of the non-parallel sides for a trapezium; only the **parallel** sides enter the formula.
- Adding instead of subtracting when an L-shaped region is obtained by removing a corner.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 120 cm²	base × height $= 15 \times 8$
2	**(b)** 80 cm²	$\frac{1}{2}(9+7) \times 10$
3	**(a)** 40 cm²	$\frac{1}{2} \times 16 \times 5$
4	**(c)** 121 cm²	Side $= 11$ cm; area $= 11^2$
5	**(a)** 140 cm²	14×10
6	**(b)** 24 cm²	$\frac{1}{2} \times 16 \times 3$
7	**(b)** 84 cm²	base × height $= 7 \times 12$
8	**(b)** 80 cm²	$\frac{1}{2}(15+5) \times 8$
9	**(b)** 36 cm²	$\frac{1}{2} \times 18 \times 4$
10	**(a)** 144 cm²	Side $= 12$ cm; area $= 12^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $= \frac{1}{2}d_1d_2 = \frac{1}{2} \times 30 \times 12$ (1) $= 180$ cm² (1).	1+1
12	Area $= l \times b = 22 \times 5$ (1) $= 110$ m² (1).	1+1
13	Area $= \frac{1}{2}bh = \frac{1}{2} \times 30 \times 8$ (1) $= 120$ cm² (1).	1+1
14	Area $= \frac{1}{2}d_1d_2 = \frac{1}{2} \times 24 \times 9$ (1) $= 108$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= bh = 150$ cm² (2); triangle $= \frac{1}{2} \times 150 = 75$ cm² (1).	2+1
16	Area $= \frac{1}{2}(a+b)h = \frac{1}{2}(24+6) \times 7$ (2) $= 105$ cm² (1).	2+1
17	Area $= 15 \times 4 = 60$ m² (2); cost $= 60 \times 35 =$ ₹ 2100 (1).	2+1
	OR: Area $= bh = 110$ cm² (2); triangle $= \frac{1}{2} \times 110 = 55$ cm² (1).	2+1
18	Area $= \frac{1}{2}(a+b)h = \frac{1}{2}(14+8) \times 12$ (2) $= 132$ cm² (1).	2+1
	OR: Area $= 14 \times 6 = 84$ m² (2); cost $= 84 \times 20 =$ ₹ 1680 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Side $= 56 \div 4 = 14$ m (1); area $= 14^2 = 196$ m² (2); tiles $= 196$ (2).	1+2+2
	OR: Area $= \frac{1}{2} \times 30 \times 8 = 120$ cm² (3); each triangle $= \frac{1}{4}$ of it $= 30$ cm² (2).	3+2
20	Side $= 64 \div 4 = 16$ m (1); area $= 16^2 = 256$ m² (2); tiles $= 256$ (2).	1+2+2
	OR: Area $= \frac{1}{2} \times 28 \times 12 = 168$ cm² (3); each triangle $= \frac{1}{4}$ of it $= 42$ cm² (2).	3+2

Mistakes to avoid:

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- Mixing units: multiplying a length in metres by one in centimetres; convert first.
- Writing area in cm instead of cm^2, or converting m^2 to cm^2 by multiplying by 100 (it is 10 000).
- Using the sum of the non-parallel sides for a trapezium; only the **parallel** sides enter the formula.
- Adding instead of subtracting when an L-shaped region is obtained by removing a corner.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.